SRM VALLIAMMAI ENGINEERING COLLEGE
Approved by AICTE | Affiliated to Anna University | Accredited by NBA
'A' Grade Accreditation by NAAC | ISO 9001 : 2015 Certified
[An Autonomous Institution]

DEPARTMENT OF CIVIL ENGINEERING

ACADEMIC YEAR: 2024-2025

LAB MANUAL

CE3368-COMPUTER AIDED DRAFTING AND MODELLING LABORATORY

2023 Regulation

III Semester B.E. Civil Engineering

LAB MANUAL

(2024-2025 ACADEMIC YEAR)

Prepared by:

1. Mr. G.R.Iyappan / AP-Civil

SRM VALLIAMMAI ENGINEERING COLLEGE

(An Autonomous Institution Affiliated to Anna University, Chennai,'A' Grade Accredited by NAAC, NBA Accredited, ISO 9001: 2015 Certified)

B.E. CIVIL ENGINEERING

REGULATIONS – 2023

1. PROGRAMME EDUCATIONAL OBJECTIVES (PEOs):

1. To produce graduates who can understand their ethical, environmental as well as professional responsibilities so that they appreciate the impact of the engineering solutions which have sustainability over society and the nation.
2. To develop the graduates who will exhibit strong technical ability to create & synthesize data using relevant tools and concepts, for providing sustainable solutions to civil engineering problems and projects.
3. To equip the graduates with suitable skills making them industry ready when they leave the portals of the Institute and to become a competent distinguished Professional Civil Engineer.
4. To produce students who can exhibit attitude, professionalism, ability to communicate with team members and adapt to the latest technology by engagingthemselves in life-long learning

2. PROGRAMME OUTCOMES (POs):

After going through the four years of study, our Civil Engineering Graduates will exhibitability to:

PO	Graduate Attribute	Programme Outcome
1	Engineering knowledge	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization for the solution of complex engineering problems.
2	Problem analysis	Identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3	Design/development of	Design solutions for complex engineering problems

	solutions	and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.
4	Conduct investigations of complex problems	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions
5	Modern tool usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling to complex engineering activities, with an understanding of the limitations.
6	The engineer and society	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice
7	Environment and sustainability	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice
9	Individual and team work	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings
10	Communication	Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions
11	Project management and finance	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments
12	Life-long learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

3. PROGRAM SPECIFIC OUTCOMES (PSOs):

By the completion of Civil Engineering program the student will have following Program specific outcomes

1. Establish a Civil Engineering career in industry, government or academic field and achieve professional expertise as appropriate.
2. Execute innovation and excellence in Civil engineering problem solving and design in global and societal contexts.
3. Commit to lifelong learning and professional development in the Civil Engineering field tostay updated in technology, research topics and contemporary issues.
4. Understand the fundamentals of Civil Engineering in commercial contexts and in expediting construction projects.

4. PEO / PO Mapping:

PROGRAMME EDUCATIONAL OBJECTIVES	PROGRAMME OUTCOMES												PROGRAM SPECIFIC OUTCOMES			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4
I	3	1	1	1	2	3	3	3	2	3	1	3	3	3	1	1
II	2	3	3	3	3	2	3	3	2	2	3	3	2	2	3	3
III	1	-	-	3	3	3	1	1	3	-	2	1	1	3	-	3
IV	3	-	-	1	3	3	3	-	3	3	2	3	3	3	-	1

Contribution: **1. Reasonable** **2. Significant** **3. Stro**

B.E. CIVIL ENGINEERING

Vision

To produce competent and quality engineers by imparting knowledge, excellence and global perspectives in Civil Engineering to our students and to make them ethically strong professional engineers to build our nation.

Mission

- To produce outstanding graduates with high technical knowledge to serve the nation.
- To impart value based education
- To provide solution to the challenges in the field of Civil Engineering.

Program Educational Objectives (PEOs)

1. To produce graduates who can understand their ethical, environmental as well as professional responsibilities so that they appreciate the impact of the engineering solutions which have sustainability over society and the nation.
2. To develop the graduates who will exhibit strong technical ability to create & synthesize data using relevant tools and concepts. for providing sustainable solutions to civil engineering problems and projects.
3. To equip the graduates with suitable skills making them industry ready when they leave the portals of the Institute and to become a competent distinguished Professional Civil Engineer.
4. To produce students who can exhibit attitude, professionalism, ability to communicate with team members and adapt to the latest technology by engaging themselves in life-long learning

INDEX

EXPT. NO	DATE	NAME OF THE EXPERIMENT	STAFF SIGN	REMARKS
1		Study of Capabilities of Software for Drafting and Modeling – Coordinate Systems		
2		Drawing of a Title Block with necessary Text and Projection Symbol.		
3		Drawing of Curves a) Parabola b) Spiral c) Involutes		
4		Drawing of Front View And Top View Of Simple Solids a) Prism b) Pyramid c) Cylinder d) Cone		
5		Drawing Front View, Top View And Side View a) V-Block b) Base Of A Mixie c) Simple Stool d) Objects With Hole And Curves		
6		Drawing of a Plan of Residential Building		
7		Drawing of a Simple Steel Truss		
8		Drawing Sectional Views a) Prism b) Pyramid c) Cylinder d) Cone		

9		Drawing Isometric Projection of Simple Objects		
10		Creation Of 3-D Models of Simple Objects		
11		Obtaining 2-D Multi-View Drawings from 3-D Model		

CE3368 COMPUTER AIDED DRAFTING AND MODELLING LABORATORY L T P C 0 0 3 1.5

COURSE OBJECTIVES:

- To develop skill to use software to create 2D and 3D models.
- To apply basic concept to drawing, edit, dimension, hatching etc., to develop 2D&3D Modelling
- To Understand and interpret the engineering drawings.
- To introduce standards and codes to produce engineering drawings.
- To provide hands on training to become proficient with 2D Computer Aideddrafting of simple objects.

LIST OF EXPERIMENTS:

1. Study of capabilities of software for Drafting and Modeling – Coordinate systems.

2. Creation of simple figures like polygon and general multi-line figures and Title Blockwith necessary text, projection and symbols.

3. Drawing of curves like parabola, spiral, involute using Bspline or cubic spline.

4. Drawing of front view and top view of simple solids like prism, pyramid, cylinder,cone, etc, and dimensioning.

5. Drawing front view, top view and side view of objects from the given pictorial views.

6. Drawing sectional views of prism, pyramid, cylinder, cone, etc

7. Drawing isometric projection of simple objects

8. Creation of 3-D models of simple objects and obtaining 2-D multi-view drawings from3-Dmodel

9. Drawing of a plan of residential building.

LIST OF EQUIPMENTS FOR A BATCH OF 30 STUDENTS:

Sl.No.	Description of Equipment	Quantity
1.	Pentium IV computer or better hardware, with suitable graphics facility	30 No.
2.	Licensed software for Drafting and Modeling.	30 Licenses
3.	Laser Printer or Plotter to print / plot drawings	2 No

TOTAL: 45 PERIODS

COURSE OUTCOMES:

1. The students will have the required knowledge in the area of 2D & 3D Views using computer aided drafting software.
2. Will be able to construct accurate 2D geometry as per the dimensions following standard drawing practices with proper dimensioning using Computer Aided drafting software.
3. Will be able to develop isometric drawings views using Computer Aided drafting software
4. Will be understand the Create 2D representations of 3D objects as plan view, elevations, side views and sections / auxiliary views using Computer Aided drafting software
5. Will be able to standard drawing codes and practices which is required for producing engineering drawings.

TEXTBOOKS:

1. Sikka V.B., A Course in Civil Engineering Drawing, 4th Edition, S.K.Kataria and Sons,2015.
2. George Omura, Mastering in Autocad 2005 and Autocad LT 2005– BPB Publications, 2008.

REFERENCES:

1. Chuck Eastman, Paul Teicholz, Rafael Sacks, Kathleen Liston, BIM Handbook: A Guide to building information modeling for Owners, Managers, Designers, Engineers, and Contractors, John Wiley and Sons, Inc.,2011.
2. Marimuthu V.M., Murugesan R. and Padmini S., Civil Engineering Drawing-I, PratheebaPublishers, 2008.
3. Vijay Duggal, "A general guide to Computer Aided Design & Drafting", Mailmax Publications, 2000.

CE3368	PROGRAM OUTCOMES												PSO			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4
CO1	3	-	-	-	3	2	-	-	2	-	-	1	2	-	1	3
CO2	3	-	-	-	3	2	-	-	2	-	-	1	2	-	1	3
CO3	3	-	-	-	3	1	-	-	2	-	-	1	2	-	1	3
CO4	3	-	-	-	3	2	-	-	2	-	-	1	2	-	1	3
CO5	3	-	-	-	3	3	-	-	2	-	-	1	2	-	1	3
Average	3.0	-	-	-	3.0	2.0	-	-	2.0	-	-	1.0	2.0	-	1.0	3.0

INTRODUCTION TO AUTOCAD

DRAWING AND MODIFYING OBJECTS

AutoCAD is drafting/modeling software used all over the world by almost all Manufacturing companies. It is variable software which can be used in all engineering divisions. It is a drafting version popularly known to everyone associated with mechanical engineering. The AutoCAD drawing enables the designer to communicate his ideas to the outside of department easily.

Conversion of AutoCAD files to other software is also using drawing exchange Formula. In the lesson, commands and procedures for drawing and modifying the Objects are explored.

START AND SAVE A DRAWING

When you start a drawing, you specify the type of units and other settings you can also choose how to save your files, including saving back up files.

The settings you select, English a metric determines default values used for many system variables of controlling text dimensions, grid, snap and default line type and hatch pattern life.

ENGLISH - Creates a new drawing based on Imperial Measurement System. The drawing is based on cad.dwt template.

METRIC

Creates a new drawing based on metric measurement. The drawing is based on Ocadiso.dwt template. Save drawing files for later use.

CONTROL THE VIEWS

You can magnify the details in your drawing for a closer view or shift the view to a different part of the drawing. If you can save the view by name, you can restore them later.

UNITS

Every object is measured in units. In AutoCAD we can determine the value of the units before we draw.

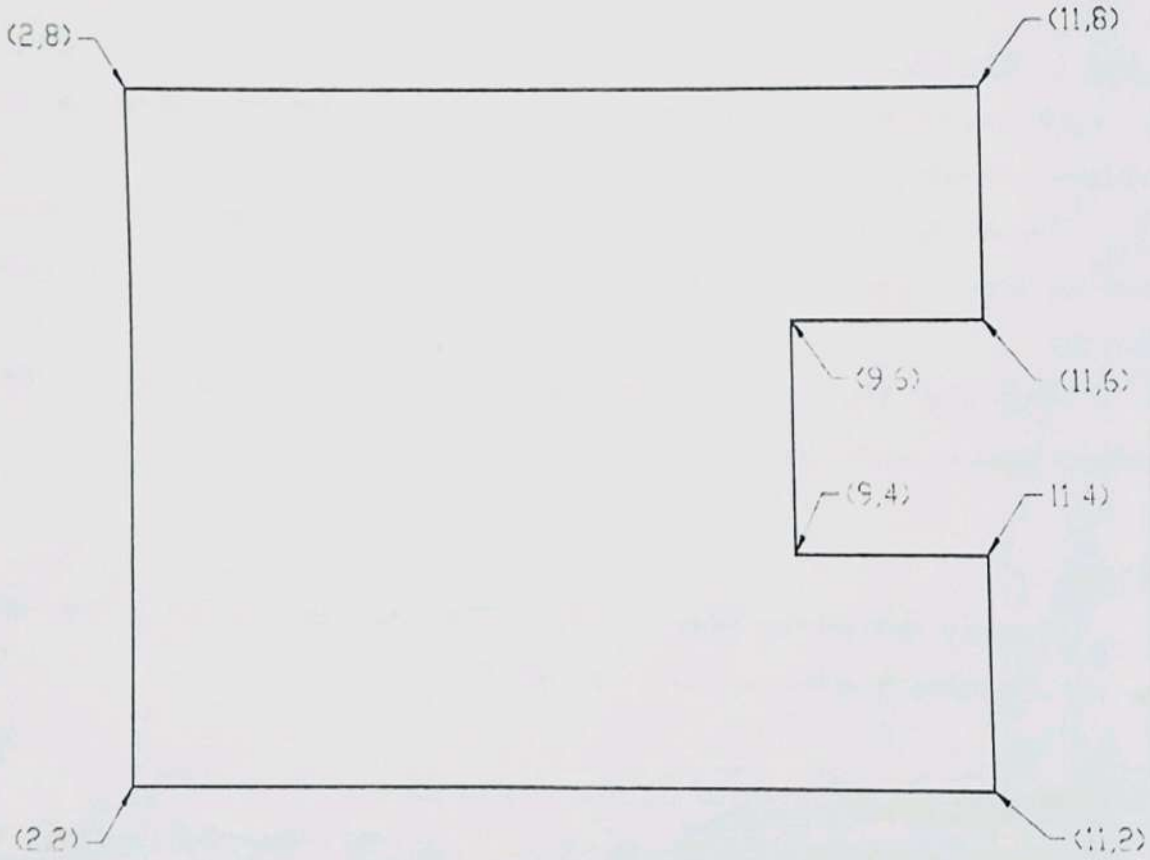
LIMITS

The drawings limits are two-dimensional points in the world coordinate that represent a lower-left limit and an upper right limit. You cannot impose limits on the Z direction.

AUTOCAD COORDINATE ENTRY METHODS

ABSOLUTE METHOD: (X,Y)

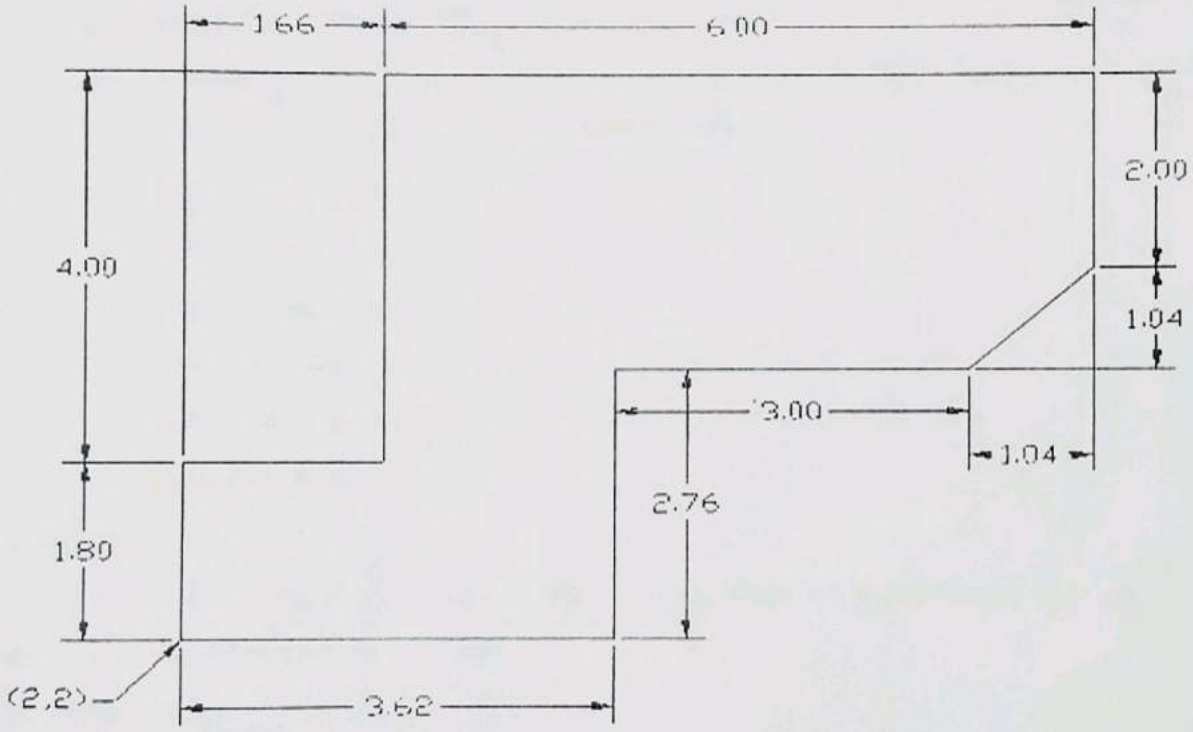
Absolute Cartesian coordinates specify a point's exact distance from the origin point of the coordinate system, which is represented as (0,0). The absolute X and Y coordinates are signed numbers.



RELATIVE METHOD: (@X,Y)

Relative Cartesian coordinates specify a point's exact distance from the last point that was entered.

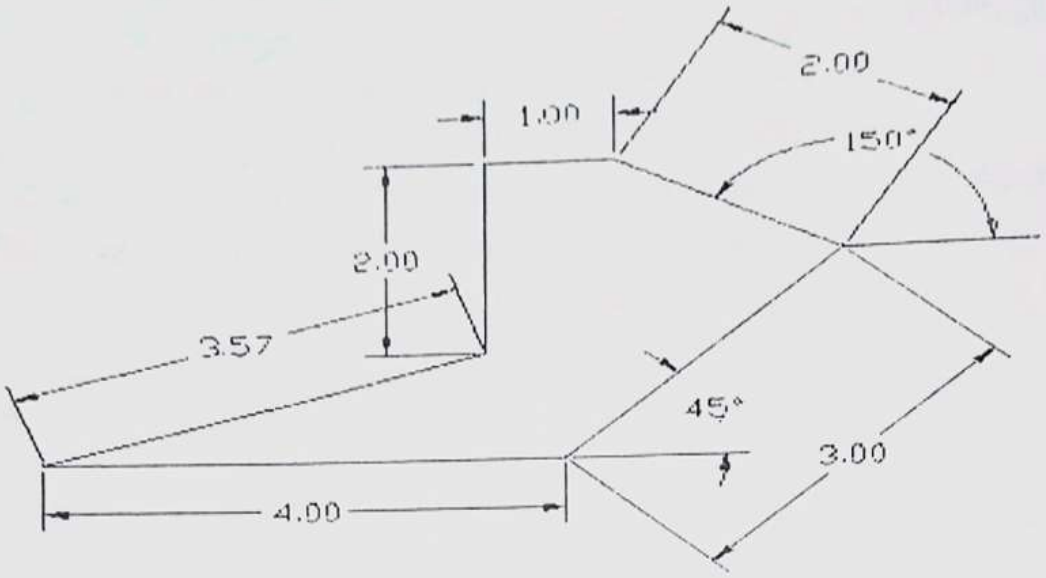
For example, typing @4,2 tells AutoCAD to locate a point that is four X units and two Y units away from the last point entered. The X and Y relative coordinates are signed numbers.Direct distance entry is a shorthand relative coordinate entry method.



POLAR METHOD: (@DISTANCE<ANGLE)

Polar coordinates specify a point's exact location by a distance and angle from the last point that was entered. The distance is always positive and the angle is measured from the positive X axis.

For example, typing @4<45 tells AutoCAD to locate a point that is four units away from the current location and at an angle of 45 degrees from the horizontal.



Polar method

AUTO CAD COMMANDS:

LINE

Line command allows you to create a line where the end points are specified by two dimensional or three dimensional coordinates.

POLYLINE

PLINE command allows you to draw line and arc segments, but from start to end, it is treated as a sing object. With PLINE command, you can draw objects even with line width.

ARC

The ARC command allows you to create an arc segment. There are different methods of creating an arc. The different methods of creating an arc are:

- 3 Points
- Centre, Start, Radius
- Start, Centre, End
- Start, Centre, Angle
- Start, Centre, Length
- Start, End, Angle
- Start, End, Direction
- Start, End, Radius

CIRCLE

The CIRCLE command allows you to create a circle. There are four different methods for drawing circles. They are:

- Centre
- Radius/Centre,
- Diameter
- 3 Points (3P)
- 2 Points (2P)
- Tangent, Tangent, Radius

POLYGON

Polygon command creates a regular polygon. You must specify the number of sides of the polygon and whether it is Inscribed or Circumscribed polygon. Now, specify the centre and radius of the polygon circle.

OSNAP

Osnap allow you to *snap* onto a specific object location when you are picking a point. For example, using Osnap you can accurately pick the end point of a line or the center of a circle.

ERASE

This command is used for deleting unwanted objects. You must select the object to be erased.

COPY

The copy command is used for making copies of selected objects. The object to be copied must be selected and the base point must be specified and the copy can be dragged and placed at the required position.

MOVE

This command is used for moving selected objects. The object to be moved must be selected and its base point must be specified then we can drag it to the required location

ROTATE

This command is used for rotating selected objects. To rotate an object, first select it and specify a base point. Now enter an angle value or specify a second point to rotate the object.

Entering a positive angle value rotates the objects counterclockwise or clockwise, depending on the Direction Control setting and Drawing Units dialog box. The plane of rotation and the direction of the zero angles depend on the orientation of the user coordinate system.

MIRROR

The MIRROR command is used for creating mirror images of selected objects. You must first select the object to be mirrored and then specify the axis along which it is mirrored, to create an mirror image of the selected object. This is a very useful command for drawing symmetrical objects about a particular axis.

OFFSET

Offset an object to create a new object whose shape parallels the shape of the original object. Offsetting a circle or an arc creates a larger or smaller circle or arc, depending on which side you specify for the offset.

ZOOM ALL

This zoom to display the entire drawing in the current viewport. The display shows all the entities even if the drawing extends outside the drawing limits.

ZOOM WINDOW

ZOOM WINDOW commands to display an area specified by two diagonally opposite corner points of a rectangle window.

ZOOM EXTENTS

This zoom to display the drawing extents. If the drawings are small in the existing limits, then there is an enlarging effect on the screen. If the drawing occupies the complete area of the limits, then there are many not being much difference between ZOOM ALL and ZOOM EXTENTS.

DIMLINEAR

This command specifies the linear distance between two selected points.

DIMALIGNED

This command is used for specifying the linear distance (exact dimensional value) between two points which are inclined at an angle.

DIMANGULAR

This command is used for obtaining the angle two selected points. The three types are:

1) Arc Selection
2) Circle Selection
3) Line Selection
4) Three Point Selection

DIMRADIUS

This command gives the value of radius of the specified circle, arc or fillet. A radial dimension consists of a radius dimension line with an arrowhead at the arc or circle end.

DIMCENTER

This command marks the centre point of the selected arc or circle. The center mark cannot be updated, so set the required style before executing this command.

LINETYPE

LINETYPE greatly improves the readability of technical drawings. You can make important features stand out with bold line weights.

LAYER

Layer is an organizing tool. Layers are like one, where you can keep various types of information. It's always a good idea to keep notes and reference symbols about each element of the drawing as well as drawing dimensions on a new layer.

DISTANCE

This command lists the distance between the selected points and also lists out the angle in current plane and 3D angle from the current plane, based on the direction of picking the two points. It also lists the projected lengths on the three axes called the delta length.

TEXT

TEXT command creates text objects with specified height and orientation. Text objects can be created with a variety of character patterns call style. This command allows you to type a single line or multiple lines of text.

MTEXT

MTEXT creates paragraph that fit within a nonprinting text boundary. The user specified text boundary determines the width of the paragraph and the justification of the text within the paragraph. Each multiline text object is a single object, regardless of the number of lines it contains.

HATCH

Hatches are shaded pattern which generally represent cross-section of a mechanical component. Using BHATCH command the required area, type of hatch, hatch angle and scale can be specified.

PROPERTIES

The PROPERTIES command displays the Properties palette. The Properties palette is the main method for viewing and modifying the properties of AutoCAD objects.

VPORTS

Viewports are areas that display different views of your model. As you work on the Model tab, you can split the drawing area into one or more adjacent rectangular views known as *model space viewports*. In large or complex drawings, displaying different views reduces the time needed to zoom or pan in a single view. Also, errors you might miss in one view may be apparent in the others.

EXTRUDE

Solids can be created by extruding selected objects. Use the EXTRUDE command to create a solid or surface from a common profile of an object.

REVOLVE

REVOLVE command can create a solid or surface by revolving open or closed objects about an axis. The revolved objects define the profile of the solid or surface.

PYRAMID

This command can create a solid pyramid. The number of sides for a pyramid, from 3 to 32, can be defined. The axis endpoint defines the length and orientation of the pyramid.

CREATE LAYOUTS

A layout stimulates a sheet of paper and provides a predictable plotting setup for a layout, you can create and position view post objects and you can add a title block or other objects geometry.

PLOT DRAWINGS

Once you have completed a drawing you can plot the drawing on paper or create a file for use with another application. In either case, you select the plot settings.

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3.	CREATION OF TITLE BLOCK
4.	CREATION OF PARABOLA
5.	CREATION OF CUBIC SPLINE
6.	CREATION OF HEXAGONAL PRISM
7.	CREATION OF PENTAGONAL PRISM
8.	CREATION OF CONE
9.	CREATION OF CYLINDER
10.	SECTIONAL VIEW OF CYLINDER
11.	SECTIONAL VIEW OF CONE
12.	RESIDENTIAL BUILDING PLAN
13.	CREATION OF STEEL TRUSS
14.	CREATION OF ORTHOGRAPHIC PROJECTIONS OF THE BASE OF MIXER
15.	CREATION OF ORTHOGRAPHIC PROJECTIONS OF V-BLOCK
16.	ISOMETRIC DRAWING
17.	3D VIEW OF CONE
18.	3D VIEW OF HEXAGONAL PRISM
19.	CREATION OF 3-D MODEL OF THE BASE OF THE MIXER

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INTRODUCTION

BASIC DRAWING COMMANDS FOR AUTOCAD

MEASURING COMMANDS			
GRID	**F7**	Displays a grid of dots at a desired spacing on the screen.	GRID (enter) On/Off/Tick spacing(x)/Aspect: (enter value) (enter)
SNAP	**F9**	Specifies a "round off" interval so that points entered with the mouse can belocked into alignment with the grid spacing.	SNAP (enter) On/Off/Value/Aspect/Rotate/Style: (enter value) (enter)
BASIC DRAWING COMMANDS			
CIRCLE	**C**	Draws circles of any size.	Circle (enter) 3P/2P/TTR/<center point>: (pick a center point) Diameter or <Radius>: (Pick a point on the circle)
LINE	**L**	Draws straight lines between two points	LINE (enter) From Point: (pick a point using the mouse) To Point: (Pick a point using the mouse) To Point: (Press return to end the command)
ARC	**A**	Draws an arc (any part of a circle or curve) through three known points.	ARC (enter) Center/ < Start point > : (pick the first point on the arc) Center/End/ < Second point > :**C** Center: (pick the arc's center point) Angle/Length of chord/ <End point > : (pick the arc endpoint)
DISPLAY COMMANDS			
LIMITS		Sets the size of the drawing paper. For size "A" drawing paper the limits should beset for 10.5 x 8.	LIMITS (enter) On/Off/Lower left corner <0.0000> (enter) Upper right corner: 10.5,8 (enter)

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ZOOM	**Z**	Enlarges or reduces the display of a drawing.	ZOOM (enter) All/Center/Dynamic/Extents/Left/Previous/V max/Window/<Scale(x/XP)>:(pick a point to define one corner of a rectangular viewing window then pick a point to define the second point to define the opposite diagonal corner of the viewing window). Note: T return the picture to its original viewing size enter **ALL** and press the Enter key when prompted instead of defining a window.
PAN	**P**	Allows us to move the view point around the drawing without changing themagnification factor.	PAN (enter)
EDITING COMMANDS			
CHANGE		Alters properties of selected objects	CHANGE(enter) Select objects or window or Last (select objects to be changed) Properties/<Change point>: (type P) Change what property (Color/Elev/ Layer/L Type/Thickness)? (Type Layer) New Layer: (enter new layer name and press enter)
ERASE	**E**	Erases entities from the drawing.	ERASE (enter) Select objects or Window or Last: (Select objects to be erased and press enter when finished)
EXTEND	**EXT**	Lengthens a line to end precisely at a **boundary edge.**	Extend (enter) Select boundary edge(s)...Select Objects (pick the line which represents the boundary edge which lines will be extended to)(press enter when finished selecting cutting edges)<Select object to extend>/Undo: (pick the line(s) that need to be extended.
TRIM	**TR**	Trims a line to end precisely at a **cutting**	Trim (enter) Select cutting edge(s)... Select Objects (pick th

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		edge.	line which represents the cutting edge of line in which objects will be trimmed to)(press enter when finished selecting cutting edges)<Select object to trim>/Undo: (pick the line(s) that need to be trimmed).
GRIPS		We can edit selected objects by manipulating grips that appear at defining points on the object. Gripsis not a command.	To activate grips simply pick the object. Small squares will appear at various entity-specific positions. By selecting an end grip we can stretch the entity to change its size. By selecting the center grip we can move the entity to a new location. To remove grips press CTL-C twice. We can perform the following using grips: Copy, Multiple Copy, Stretch, Move, Rotate, Scale, and Mirror.
CREATING LAYERS			
LAYER		Creates named drawing layers and assigns color and line type properties to those layers.	A Layer & Line type Properties dialog box will be displayed. To add a new layer, pick the new button. A new layer listing appears, using a default name of Layer1. The layer name can be changed by highlighting the layer name. Colors and Line types can be assigned to each new layer by picking the color box to assign a color and picking the line type box to assign a line type.
CONSTRUCTION COMMANDS			
ARRAY	**AR**	Makes multiple copies of selected objects in a rectangular or circular pattern	ARRAY (enter) Select objects or Window or Last: (select object to array) Rectangular or Polar array (R/P) <current>: (P) Center point of array: (pick the point around which to form the array) Angle to fill (+=CCW, -=Cw) <360>: (enter)
COPY	**CO**	Draws a copy of selected objects.	COPY (enter) Select objects or Window or Last: (select objects to be copied) Base point or displacement: (pick a point on the object to be use as a reference point) Second point of displacement: (pick a point which represents the new location of the

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			copied object)
MIRROR	**MI**	Makes mirror images of existing objects.	MIRROR (enter) Select objects or Window or Last: (select objects to be mirrored) First point of mirror line: (pick a point on top of the mirror line) Second point: (pick a point on the bottom of the mirror line) Delete old objects? <N> y or n (enter)
MOVE	**M**	Moves designated entities to another location.	MOVE (enter) Select objects or Window or Last: (select objects to move) Base point or displacement: (pick a point on the object to be use as a reference point) Second point of displacement: (pick a point which represents the new location of the object)
OFFSET	**O**	Constructs an entity parallel to another entity at a specified distance. Offset canbe used with lines, circles, arcs, and polylines.	OFFSET (enter) Offset distance or Through<last>: (enter a distance value) Select object to offset: (select object to offset) Side to offset: (Pick any point on the side of the object we wish to offset)
FILLET	**FI**	Changes any corner to a rounded corner.	FILLET (enter) Polyline/Radius/Angle/Trim/Method/ <Select first line > : (pick the first line) Select second line: (pick the second line)
CHAMFER	**CHA**	Changes any corner to an angled corner.	CHAMFER Polyline/Distance/Angle/Trim/Method/ < Select first line > : (pick the first line) Select second line: (pick the second line)
OSNAP	**F3**	Instantly locates exact points relative to existing objects (points).	**Object Snap Modes:** Endpoint, Midpoint, Center, Quadrant, Intersection, Insertion, Perpendicular, Tangent, Nearest, Node, and None.

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LETTERING COMMANDS			
TEXT	**M TEXT**	Draws text characters of any size.	TEXT (enter) Justify/Style/<Start point> : (pick a starting point or enter a justification letter) Height (0) (enter the height of the lettering) Rotating Angle (0) (enter) Text: (Enter the desired lettering) (enter)
HATCHING A DRAWING			
BHATCH		Allows the user to crosshatchareas of a section view	BHATCH (enter) The Boundary Hatch dialogue Box will be displayed. Select the Hatch Options box. The Hatch Options box will be displayed. Select the Patterns box. The Choose Hatch Pattern box will be displayed. Select the desired hatch pattern. The Hatch Options box will be displayed again. We can select a scale and rotation angle for the crosshatch pattern. Select the OK box when finished. The Boundary Hatch dialogue box will be displayed again. Select the Pick Points box. When prompted select the internal point of the are to be crosshatched. Press the enter key when finished. The Boundary Hatch dialog box will be displayed again. Select the Apply box to add the crosshatching to the drawing.

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EX.NO: 01 **CO ORDINATE SYSTEMS**
DATE: **(ABSOLUTE, RELATIVE, POLAR)**

AIM: Creations of simple block by using the three different coordinate systems.

Entering Points in AutoCAD

We can enter points directly on the command line using three different systems. The one we use will depend on which is more applicable for the situation. The three systems are as follows:

ABSOLUTE CO-ORDINATES - Using this method, we enter the points as they relate to the origin of the WCS. To enter a point just enters in the exact point as X, Y.

RELATIVE CO-ORDINATES - This allows us to enter points in relation to the first point we have entered. After we've entered one point, the next would be entered as @X, Y. This means that AutoCAD will draw a line from the first point to another point X units over and Y units up relative to the previous point.

POLAR CO-ORDINATES - We would use this system if we know that we want to draw a line a certain distance at a particular angle. We would enter this as @D<A. In this case, D is the distance and A is the angle. Example: @10<90 will draw a line 10 units straight up from the first point. The three ways of entering co-ordinates shown above are the ONLY way AutoCAD accepts input. First decide which style we need to use, and then enter as shown. Remember that X is always before Y (alphabetical). Don't forget the '@' symbol when we are entering relative points.

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RESULT:

Thus the Simple Block Drawing Was Created Successfully Using Auto Cad Software by giving three different co ordinate systems.

EX.NO: 02 **CREATION OF SIMPLE BLOCK**
DATE:

AIM: Creations of simple block by using the commands of circle, point, trim,mirror, etc

Command: limits
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000, 0.0000>: 0, 0
Specify upper right corner <12.0000, 9.0000>: 300,300

Command: zoom
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real-time>:
all Regenerating model.

Command: line
Specify first point: 110, 50
Specify next point or [Undo]: 260, 50
Specify next point or [Undo]: 260,180
Specify next point or [Close/Undo]: 110,180
Specify next point or [Close/Undo]: close

Command: point
Specify the point: 185,180

Command: _circle
Specify center point for circle or [3P/2P/Ttr (tan tan /radius)]: 185,180
Specify radius of circle or [Diameter]: 35

Command: arc
Specify start point of arc or [Center]: ce
Specify center point of arc: @45<0
Specify end point of arc or [Angle/chord Length]: A
Specify included angle: 180

Command: Trim
Select the object: pick the source object
Select the object to trim: trim unwanted line

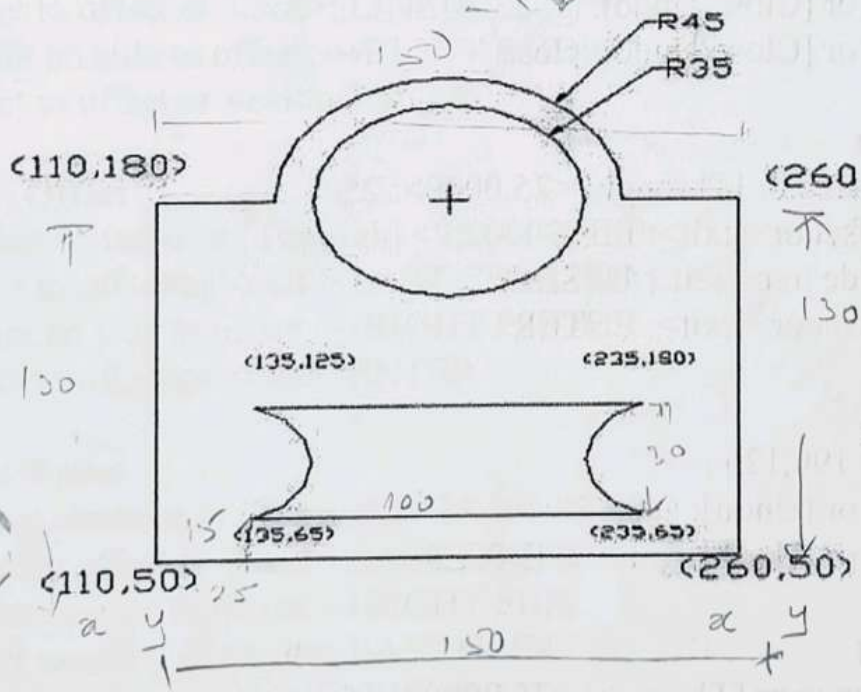
VEL TECH VEL IAMMAL ENGINEERING COLLEGE

Command: line
Specify first point: 135,125
Specify next point or [Undo]: 235,135 ENTER

Command: point
Specify the point: 235, 95

Command: arc
Specify start point of arc or [Center]: ce
Specify center point of arc: (pick the start point) (110, 50)
Specify end point of arc or [Angle/chord Length]: (pick the end point) (235,125)

Command: mirror
Select the object: pick the arc
Specify first point of mirror line: pick the midpoint of line 1
Specify next point of mirror line: pick the mid-point of line2
Delete the source objects :{ yes/no}: no ENTER



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RESULT:
Thus the Simple Block Drawing Was Created Successfully Using Auto CadSoftware.

EX.NO: 03 **CREATION OF TITLE BLOCK**
DATE:

Aim : To draw the following title block with necessary text and projection symbol using Auto CAD Software.

Command: limits
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: 0,0
Specify upper right corner <12.0000,9.0000>: 500,500

Command: zoom
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: all
Regenerating model

Command: LINE
Specify first point: 100,150
Specify next point or [Undo]: @230,0
Specify next point or [Undo]: @0,130
Specify next point or [Close/Undo]: @-230,0
Specify next point or [Close/Undo]: close

Command: Offset
Specify offset distance or [Through] <25.0000>: 25
Select object to offset or <exit>: LINE 1
Specify point on side to offset: ↑ UPSIDE
Select object to offset or <exit>: ENTER

Command: LINE
Specify first point: 190,125
Specify next point or [Undo]: @0,45
Specify next point or [Undo]: @140,0 ENTER

Command: Offset
Specify offset distance or [Through] <25.0000>: 15
Select object to offset or <exit>: LINE 2
Specify point on side to offset: ↑ UPSIDE
Select object to offset or <exit>: LINE 3
Specify point on side to offset: ↑ UPSIDE
Select object to offset or <exit>: LINE 3

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Specify point on side to offset: ↑ UPSIDE
Select object to offset or <exit>: LAST CRAEATED LINE ENTER

Command: LINE
Specify first point: 190,170
Specify next point or [Undo]: @0,60 ENTER

Command: Offset
Specify offset distance or [Through] <25.0000>: 15
Select object to offset or <exit>: LINE A
Specify point on side to offset: ↑ UPSIDE
Select object to offset or <exit>: LAST CREATED LINE
Specify point on side to offset: ↑ UPSIDE
Select object to offset or <exit>: LAST CREATED LINE
Specify point on side to offset: ↑ UPSIDE
Select object to offset or <exit>: ENTER

Command: Offset
Specify offset distance or [Through] <25.0000>: 20
Select object to offset or <exit>: LINE B
Specify point on side to offset: →RIGHT PSIDE
Select object to offset or <exit>: ENTER

Command: Offset
Specify offset distance or [Through] <25.0000>: 50
Select object to offset or <exit>: LAST CREATED LINE
Specify point on side to offset: →RIGHT PSIDE
Select object to offset or <exit>: ENTER

Command: Offset
Specify offset distance or [Through] <25.0000>: 35
Select object to offset or <exit>: LAST CREATED LINE
Specify point on side to offset: →RIGHT SIDE
Select object to offset or <exit>: LAST CREATED LINE
Specify point on side to offset: →RIGHT SIDE
Select object to offset or <exit>: ENTER

CREATING THE TEXT
Command: Text
Current text style: "Standard" Text height: 0.2000

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Specify start point of text or [Justify/Style]: 105,118
Specify height <0.2000>: 3.5
Specify rotation angle of text <0>: 0
Enter text: TITLE

Command: Text
Current text style: "Standard" Text height: 0.2000
Specify start point of text or [Justify/Style]: 115,105
Specify height <0.2000>: 7.5
Specify rotation angle of text <0>: 0
Enter text: LAYOUT

Command: Text
Current text style: "Standard" Text height: 0.2000
Specify start point of text or [Justify/Style]: 225,105
Specify height <0.2000>: 7.5
Specify rotation angle of text <0>: 0
Enter text: NAME

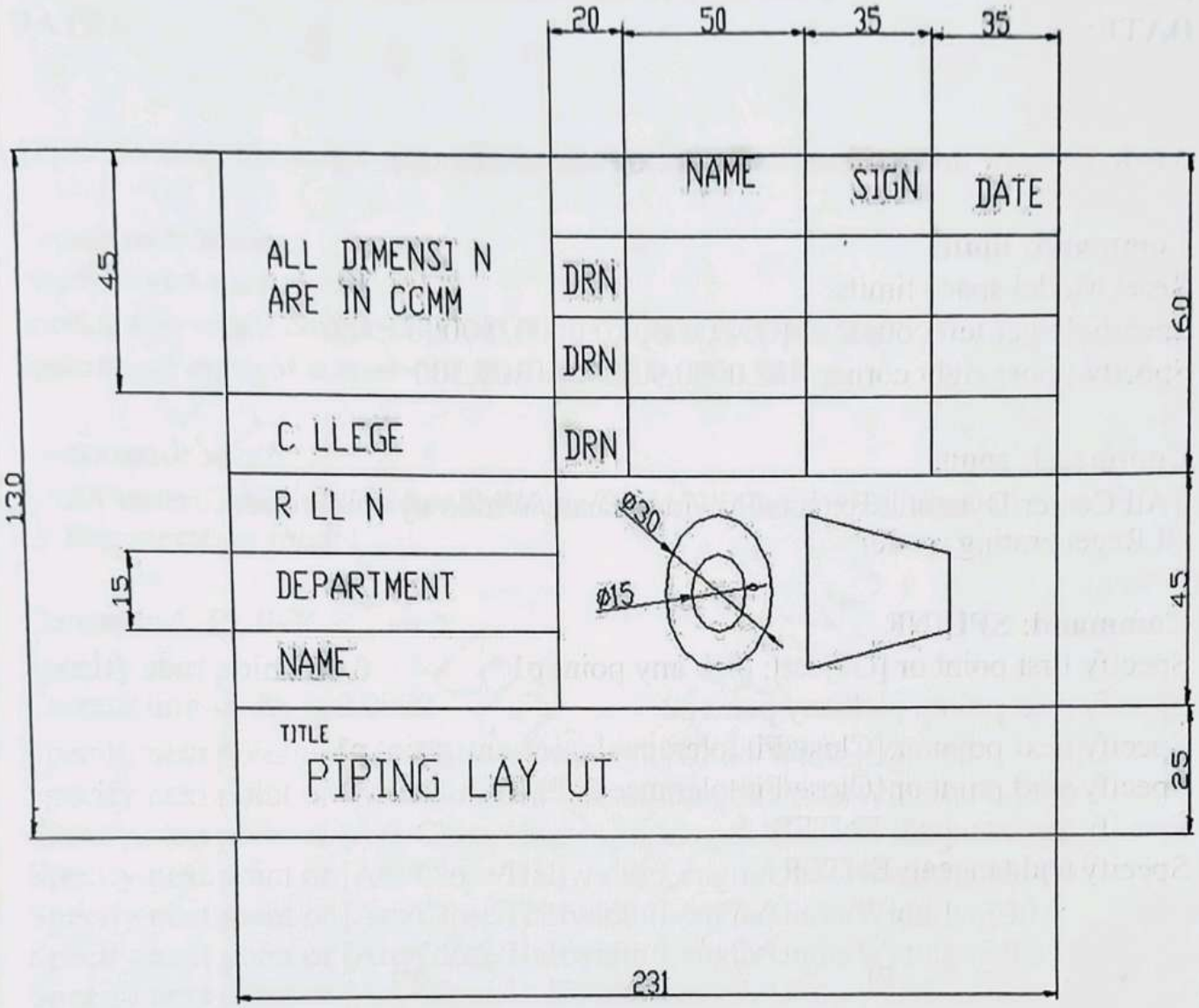
CREATING THE PROJECTION SYMBOL

Command: circle
Specify center point for circle or [3P/2P/Ttr (tan tan radius)]: 235,147.5
Specify radius of circle or [Diameter] <25.0000>: D
Specify diameter of circle <50.0000>: 30

Command: circle
Specify center point for circle or [3P/2P/Ttr (tan tan radius)]: 235,147.5
Specify radius of circle or [Diameter] <15.0000>: D
Specify diameter of circle <30.0000>: 15 ENTER

Command: LINE
Specify first point: 260,162.5
Specify next point or [Undo]: @0,-30
Specify next point or [Undo]: @40,7.5
Specify next point or [Close/Undo]: @0,15
Specify next point or [Close/Undo]: @-40,7.54
Specify next point or [Close/Undo]: Close

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TITLE BL CK

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RESULT:

Thus the following title block drawing was created successfully usingAuto Cad Software.

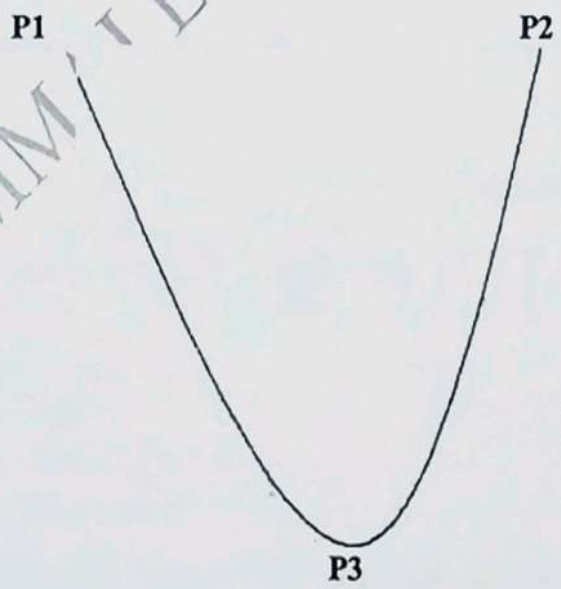
EX.NO: 04 **CREATION OF PARABOLA**
DATE:

AIM: To draw the parabola using Auto CAD Software

Command: limits
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: 0,0
Specify upper right corner <12.0000,9.0000>: 300,300

Command: zoom
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>:
all Regenerating model

Command: SPLINE
Specify first point or [Object]: pick any point p1
Specify next point : pick any point p2
Specify next point or [Close/Fit tolerance]: pick any point p3
Specify next point or [Close/Fit tolerance] : ENTER
Specify start tangent: ENTER
Specify end tangent: ENTER



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RESULT:

Thus the cubic spline drawing was successfully created using Auto CAD Software

EX.NO: 05 **CREATION OF CUBIC SPLINE**
DATE:

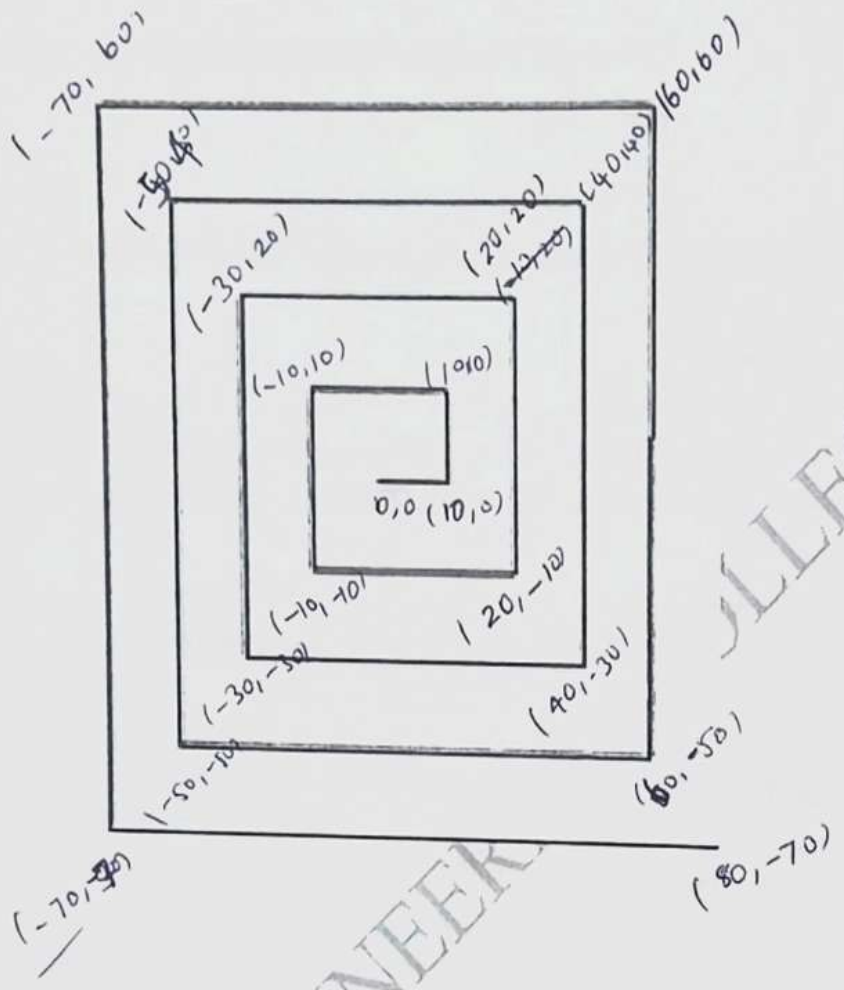
AIM: To draw the cubic spline using Auto CAD Software

Command: limits
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: 0,0
Specify upper right corner <12.0000,9.0000>: 300,300

Command: zoom
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>:
all Regenerating model

Command: PLINE
Specify start point: 10,0
Current line-width is 0.0000
Specify next point or [Arc/Halfwidth/Length/Undo/Width]: @0,10
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @-20,0
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @0,-20
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @0,-20
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @30,0
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @0,30
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @-40,0
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @0,-40
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @50,0
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @0,50
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @-60,0
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @60,0
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @0,-60 0,-60
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @70,0
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @0,70
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @-80,0
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @0,-80
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: @90,0
Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: ENTER

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CUBIC SPLINE

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RESULT:

Thus the cubic spline drawing was successfully created using Auto CAD Software.

EX.NO: 06 **CREATION OF HEXAGONAL PRISM**
DATE:

Aim : To draw the front view and top view of hexagonal prism height of 70 mmand edge 35 mm using Auto CAD Software

Command: limits
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: 0,0
Specify upper right corner <12.0000,9.0000>: 300,300

Command: zoom
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>:
all Regenerating model

FRONT VIEW:

Command: LINE
Specify first point: pick the point with respect to (a)
Specify next point or [Undo]: point with respect to (b)
Specify next point or [Undo]: 70
Specify next point or [Undo]: 70
Specify next point or [Close/Undo]: close

TOP VIEW

Command: polygon
Enter number of sides <4>: 6
Specify center of polygon or [Edge]: e
Specify first endpoint of edge: Specify second endpoint of edge: 35 enter

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RESULT:

Thus the front view and top view of a hexagonal prism Drawing Was Created Successfully Using Auto Cad Software.

EX.NO: 07 **CREATION OF PENTAGONAL PRISM**
DATE:

Aim : To draw the front view and top view of pentagonal prism height of 70 mmand edge 35 mm using Auto CAD Software

Command: limits
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000, 0.0000>: 0,0
Specify upper right corner <12.0000, 9.0000>: 300,300

Command: zoom
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>:
all Regenerating model

FRONT VIEW:

Command: Line
Specify first point: pick the point with respect to (a)
Specify next point or [Undo]: point with respect to (b)
Specify next point or [Undo]: 70
Specify next point or [Undo]: 70
Specify next point or [Close/Undo]: close

TOP VIEW

Command: Polygon
Enter number of sides <4>: 5
Specify center of polygon or [Edge]: e
Specify first endpoint of edge: Specify second endpoint of edge: 35 enter

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RESULT:

Thus the front view and top view of pentagonal prism Drawing Was Created Successfully Using Auto Cad Software.

EX.NO: 08 **CREATION OF CONE**
DATE:

Aim : To create front view and top view of cone using Auto CAD Software

Command: limits
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: 0,0
Specify upper right corner <12.0000,9.0000>: 300,300

Command: zoom
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>:
all Regenerating model.

FRONT VIEW:

Command: Line
Specify first point: pick any point
Specify next point or [Undo]: 50
Specify next point or [Undo]: @75<108
Specify next point or [Close/Undo]: close

TOP VIEW

Command: circle
Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:pick any point
Specify radius of circle or [Diameter] <35.0000>: D
Specify diameter of circle <70.0000>: 50 ENTER

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RESULT:
Thus the cone Drawing Was Created Successfully Using Auto Cad Software.

EX.NO: 09 **CREATION OF CYLINDER**
DATE:

Aim : To create front view and top view of cylinder using Auto CAD Software

Command: limits
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: 0,0
Specify upper right corner <12.0000,9.0000>: 300,300

Command: zoom
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>:
all Regenerating model.

FRONT VIEW:

Command: LINE
Specify first point: pick any point
Specify next point or [Undo]: 40
Specify next point or [Undo]: 75
Specify next point or [Close/Undo]: close

TOP VIEW

Command: circle
Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:pick any point Specify radius of circle or [Diameter] <35.0000>: D
Specify diameter of circle <70.0000>: 40 ENTER

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RESULT:

Thus the front view and top view of a cylinder Drawing Was Created Successfully Using Auto Cad Software

EX.NO: 10 **SECTIONAL VIEW OF CYLINDER**
DATE:

AIM: To draw the sectional view of cylinder using Auto CAD Software

Command: limits
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: 0,0
Specify upper right corner <12.0000,9.0000>: 300,300

Command: zoom
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>:
all Regenerating model

Command: Circle
Specify center point for circle or [3P/2P/Ttr (tan tan radius)]: pick any point Specify radius of circle or [Diameter] : D
Specify diameter of circle <70.0000>: 50 ENTER

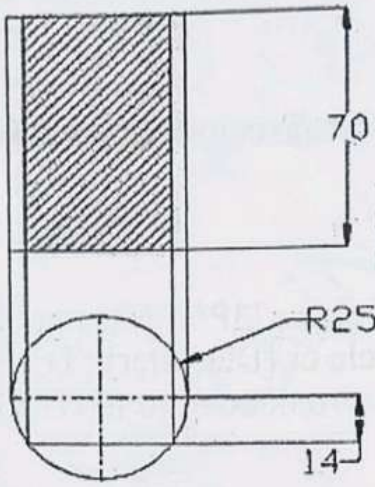
Command: Line
Specify first point: pick any point with respect to circle
Specify next point or [Undo]: @50<0
Specify next point or [Undo]: @70<90
Specify next point or [Close/Undo]: @50<180
Specify next point or [Close/Undo]: Close

Command: Offset
Specify offset distance or [Through] <25.0000>:14
Select object to offset or <exit>: LINE 1
Specify point on side to offset: ↑ UPSIDE
Select object to offset or <exit>: ENTER

Command: Line
Specify first point: pick any point with respect to offset line
Specify next point or [Undo]: both corners ENTER

Command: Hatch
Command: hatch
Select internal point: pick points ENTER

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SECTI NALVIEW F CYLINDER

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RESULT:

Thus the sectional view of cylinder was successfully created using Auto CAD Software.

EX.NO: 11
DATE:

SECTIONAL VIEW OF CONE

AIM: To draw the sectional view of cone using Auto CAD

Command: limits
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: 0,0
Specify upper right corner <12.0000,9.0000>: 300,300

Command: zoom
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>:
all Regenerating model

Command: Circle
Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:pick any point Specify radius of circle or [Diameter] : D
Specify diameter of circle <70.0000>: 35 ENTER

Command: Line
Specify first point: pick any point with respect to circle
Specify next point or [Undo]: @35<0
Specify next point or [Undo]: @42<108
Specify next point or [Close/Undo]: Close

Command: Offset
Specify offset distance or [Through] <25.0000>:14
Select object to offset or <exit>: LINE 1
Specify point on side to offset: ↑ UPSIDE
Select object to offset or <exit>: ENTER

Command: Line
Specify first point: pick any point with respect to offset line
Specify next point or [Undo]: both corners ENTER

Command: Circle
By using three points method find the diameter value and draw the circle

Command: Hatch
Command: hatch
Select internal point: pick points ENTER

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RESULT:

Thus the sectional view of cylinder was successfully created using Auto CAD Software.

EX.NO: 12 **RESIDENTIAL BUILDING PLAN**
DATE:

AIM:
To create the plane of residential building plan using AutoCAD

Command: LIMITS
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: 0,0
Specify upper right corner <420.0000,297.0000>: 500,500

Command: ZOOM
Specify corner of window, enter a scale factor (nX or nXP), or
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>:
ALL Regenerating model.

Command: line
Specify first point: @60,150
Specify next point: @90,0
Specify next point: @0,-110
Specify next point: @-46,0
Specify next point: @-30,0
Specify next point: @-14,10
Specify next point or [Close/Undo]: close

Command: line
Specify first point: @61,149
Specify next point: @88,0
Specify next point: @0,-108
Specify next point: @-44,0
Specify next point: @-28,0
Specify next point: @-0,8
Specify next point or [Close/Undo]: close

<u>To draw W-C block</u>
Command: line
Specify first point: @63,147
Specify next point: @12,0
Specify next point: @0,-10

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Specify next point: @-12,0
Specify next point or [Close/Undo]: close

To draw both block
Command: line
Specify first point: @63,136
Specify next point: @12, 0
Specify next point: @0,-18
Specify next point: @-12,0
Specify next point or [Close/Undo]: close

To draw passage block
Command: line
Specify first point: @77,147
Specify next point: @38,0
Specify next point: @0,-28
Specify next point: @-38,0
Specify next point or [Close/Undo]: close

To draw kitchen block
Command: line
Specify first point: @147,147
Specify next point: @0,-2
Specify next point: @-31,0
Specify next point: @-12,0
Specify next point or [Close/Undo]: close

To draw living room block
Command: line
Specify first point: @63,116
Specify next point: @42,0
Specify next point: @0,-58
Specify next point: @-42,0
Specify next point or [Close/Undo]: close

To draw the windows-I
Command: line
Specify first point: @63,64
Specify next point: @-9,0
Specify next point: @0,15

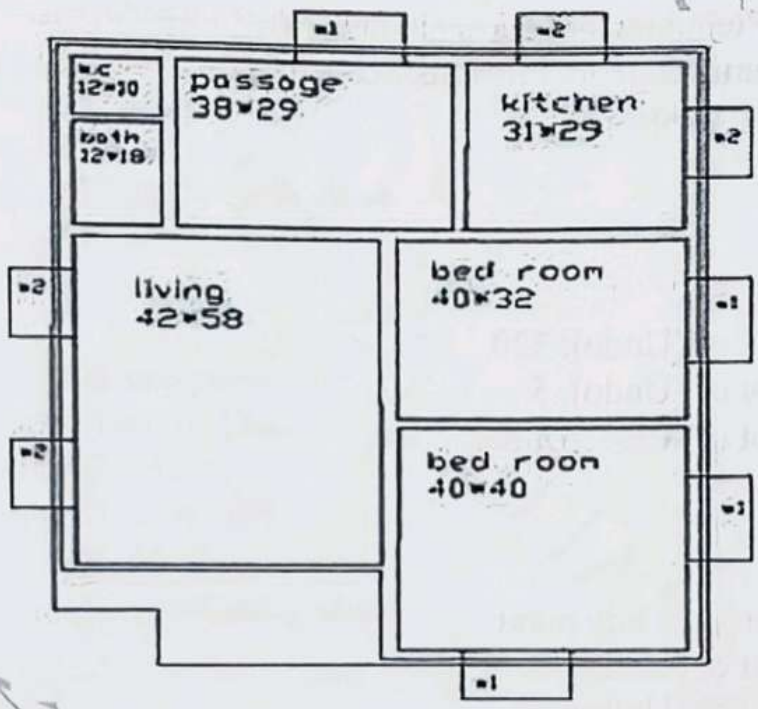
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Specify next point: @9,0
Specify next point or [Close/Undo]: close

To draw the windows-II

Command: line

Specify first point: @126,147
Specify next point: @0,9
Specify next point: @12,0
Specify next point: @9,0
Specify next point or [Close/Undo]: close



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RESULT:

Thus the plane of the residential building is successfully created using AutoCAD software.

EX.NO: 13 **CREATION OF STEEL TRUSS**
DATE:

AIM:

To create the steel truss using AutoCAD.

Command: LIMITS
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: 0,0
Specify upper right corner <420.0000,297.0000>: 500,500

Command: ZOOM
Specify corner of window, enter a scale factor (nX or nXP), or
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>:
ALL Regenerating model.

To draw B1
Command: line
Specify first point:
Specify next point or [Undo]: 320
Specify next point or [Undo]: 5
Specify next point or [Close/Undo]: close

To draw C1
Command: line
Specify first point: pick any point
Specify next point or [Undo]: 3
Specify next point or [Undo]: 65
Specify next point or [Undo]: 3
Specify next point or [Undo]: 65
Specify next point or [Close/Undo]: close

To draw C2
Command: line
Specify first point: pick any point
Specify next point or [Undo]: 3
Specify next point or [Undo]: 50
Specify next point or [Undo]: 3
Specify next point or [Undo]: 50

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Specify next point or [Close/Undo]: close

To draw C3

Command: line

Specify first point: pick any point
Specify next point or [Undo]: 3
Specify next point or [Undo]: 32
Specify next point or [Undo]: 3
Specify next point or [Undo]: 32
Specify next point or [Close/Undo]: close

To draw C4

Command: line

Specify first point: pick any point
Specify next point or [Undo]: 3
Specify next point or [Undo]: 15
Specify next point or [Undo]: 3
Specify next point or [Undo]: 15
Specify next point or [Close/Undo]: close

To draw T1

Command: line

Specify first point: pick any point
Specify next point or [Undo]: 182
Specify next point or [Undo]: 3
Specify next point or [Undo]: 182
Specify next point or [Undo]: 3
Specify next point or [Close/Undo]: close

Command: rotate

Select objects: select T1
Specify base point: pick the point C
Specify rotation angle or [Reference]: 18
Specify next point or [Close/Undo]: close

To draw S1

Command: line

Specify first point: pick any point
Specify next point or [Undo]: 66
Specify next point or [Undo]: 3
Specify next point or [Undo]: 66

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Specify next point or [Undo]: 3
Specify next point or [Close/Undo]: close

Command: rotate
Select objects: select S1
Specify base point: pick right side
Specify rotation angle or [Reference]: - 46
Specify next point or [Close/Undo]: close

To draw S2
Command: line
Specify first point: pick any point
Specify next point or [Undo]: 60
Specify next point or [Undo]: 3
Specify next point or [Undo]: 60
Specify next point or [Undo]: 3
Specify next point or [Close/Undo]: close

Command: rotate
Select objects: select S2
Specify base point: pick right side
Specify rotation angle or [Reference]: - 31
Specify next point or [Close/Undo]: close

To draw S3
Command: line
Specify first point: pick any point
Specify next point or [Undo]: 54
Specify next point or [Undo]: 3
Specify next point or [Undo]: 54
Specify next point or [Undo]: 3
Specify next point or [Close/Undo]: close

Command: rotate
Select objects: select S3
Specify base point: pick right side
Specify rotation angle or [Reference]: - 14
Specify next point or [Close/Undo]: close

To connect C1 to B1

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Command: C1 to B1
Select object: C1
Specify base point: C1(corner point)
Specify second point: B1(point)

To connect T1 to C1
Command: move
Select object: C1
Specify base point: T1(corner point)
Specify second point: C1(corner point

Mirroring the object
Command: mirror
Select objects: select all
Specify first point of mirror line: select line (point)
Specify second point of mirror line: select line (point)
Delete source objects? [Yes/No] <N>: no

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RESULT:

Thus the steel truss is perfectly created using AutoCAD software.

EX.NO: 14 **CREATION OF ORTHOGRAPHIC PROJECTIONS OF THE BASE OF MIXER**
DATE :

AIM: To draw the 3d view of base of mixer using Auto CAD Software

Command: LIMITS
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: 0,0
Specify upper right corner <420.0000,297.0000>: 500,500

Command: ZOOM
Specify corner of window, enter a scale factor (nX or nXP), or
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>:
ALL Regenerating model.

To draw the top view

Command: line
Specify first point: pick any point
Specify next point or [Undo]: 300
Specify next point or [Undo]: 150
Specify next point or [Undo]: 300
Specify next point or [Undo]: 150
Specify next point or [Close/Undo]: close

Command: Offset
Specify offset distance or [Through] <25.0000>:150
Select object to offset or <exit>: LINE 1
Specify point on side to offset: LEFT SIDE
Select object to offset or <exit>: ENTER

Command: Circle
Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:pick centre point
Specify radius of circle or [Diameter] : D
Specify diameter of circle <70.0000>: 75 ENTER

To draw the Front view

Command: Osnap
Osnap Dialogue box opens. Click "SELECT ALL".

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Command: line
Specify first point: pick any point
Specify next point or [Undo]: 300
Specify next point or [Undo]: 250
Specify next point or [Undo]: 300
Specify next point or [Undo]: 250
Specify next point or [Close/Undo]: close

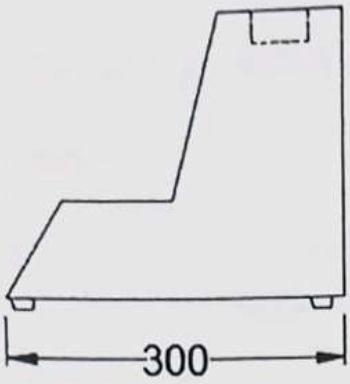
To draw the Side view
Command: Osnap
Osnap Dialogue box opens. Click "SELECT ALL".

Command: line
Specify first point: pick any point
Specify next point or [Undo]: 250
Specify next point or [Undo]: 150
Specify next point or [Undo]: 250
Specify next point or [Undo]: 150
Specify next point or [Close/Undo]: close

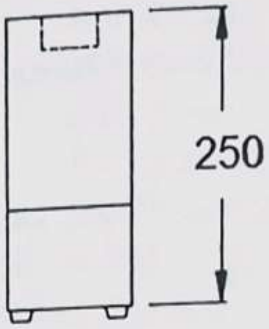
Command :Dimlinear
Specify first extension line origin or <select object> : pick start point of line for dimensioning
Specify Second extension line origin : pick the end point of the line
Drag the cursor at a suitable distance away from the drawing and left click.

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FRONT VIEW



SIDE VIEW



TOP VIEW



All dimensions are in 'mm'

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RESULT:

Thus the orthographic views of the base of mixer drawing was successfully created using Auto CAD Software.

EX.NO: 15
DATE :

CREATION OF ORTHOGRAPHIC PROJECTIONS OF V BLOCK

AIM: To draw the 3d view of v block using Auto CAD Software

Command: LIMITS
Reset Model space limits:
Specify lower left corner or [ON/OFF] <0.0000,0.0000>: 0,0
Specify upper right corner <420.0000,297.0000>: 500,500

Command: ZOOM
Specify corner of window, enter a scale factor (nX or nXP), or
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>:
ALL Regenerating model.

To draw the top view

Command: line
Specify first point: pick any point
Specify next point or [Undo]: 70
Specify next point or [Undo]: 75
Specify next point or [Undo]: 70
Specify next point or [Undo]: 75
Specify next point or [Close/Undo]: close

Command: Offset
Specify offset distance or [Through] <25.0000>:150
Select object to offset or <exit>: LINE 1
Specify point on side to offset: LEFT SIDE
Select object to offset or <exit>: ENTER

Command: Circle
Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:pick centre point
Specify radius of circle or [Diameter] : D
Specify diameter of circle <70.0000>: 75 ENTER

To draw the Front view

Command: Osnap
Osnap Dialogue box opens. Click "SELECT ALL".

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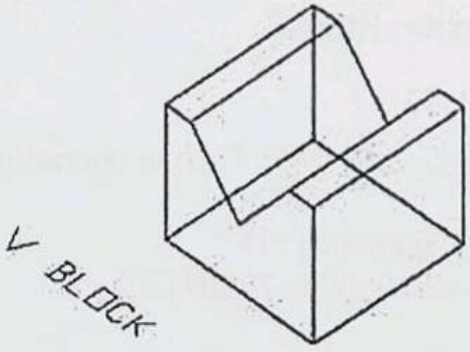
Command: line
Specify first point: pick any point
Specify next point or [Undo]: 62.5
Specify next point or [Undo]: 75
Specify next point or [Undo]: 62.5
Specify next point or [Undo]: 75
Specify next point or [Close/Undo]: close

To draw the Side view

Command: Osnap
Osnap Dialogue box opens. Click "SELECT ALL".

Command: line
Specify first point: pick any point
Specify next point or [Undo]: 75
Specify next point or [Undo]: 70
Specify next point or [Undo]: 75
Specify next point or [Undo]: 70
Specify next point or [Close/Undo]: close

Command :Dimlinear
Specify first extension line origin or <select object> : pick start point of line for dimensioning
Specify Second extension line origin : pick the end point of the line
Drag the cursor at a suitable distance away from the drawing and left click.



V BLOCK

RESULT:

Thus the 3D view of V block drawing was successfully created using Auto CAD Software.

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EX.NO: 16 ISOMETRIC DRAWING

DATE:

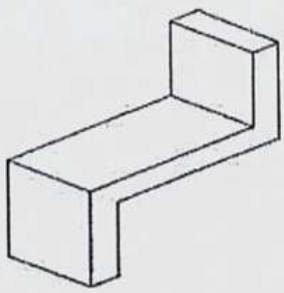
AIM: To Draw the Isometric Drawing Using Auto CAD Software

Command: snap
Specify snap spacing or [ON/OFF/Aspect/Rotate/Style/Type] <0.5000>: style
Enter snap grid style [Standard/Isometric] <S>: isometric
Specify vertical spacing <0.5000>: ENTER

Command: line
Specify first point:
Specify next point or [Undo]: 90
Specify next point or [Undo]: 120
Specify next point or [Close/Undo]: 70
Specify next point or [Close/Undo]: 20
Specify next point or [Close/Undo]: 90
Specify next point or [Close/Undo]: 120
Specify next point or [Close/Undo]: 70
Specify next point or [Close/Undo]: close

Command: line
Specify first point: pick any point
Specify next point or [Undo]: 60

Command: copy
Select the object: select the line
Specify base of displacement: pick the point ENTER



RESULT:

Thus the isometric object drawing was successfully created using Auto CAD Software.

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EX.NO: 17 3D VIEW OF CONE
DATE :

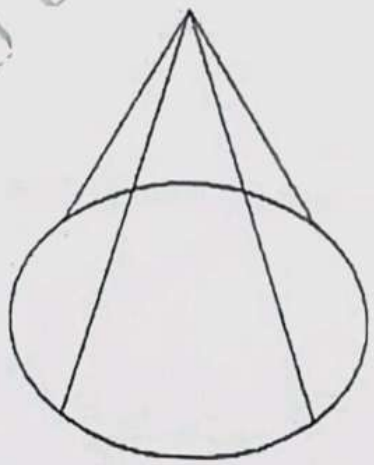
AIM: To draw the 3d view of cone using Auto CAD Software

Command: vpoint
Specify a view point or [Rotate] <display compass and tripod>: 1,-1,1
Regenerating model.

Command: zoom
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>:
all Regenerating model

Command: shademode
Enter option [2D wireframe/3D
wireframe/Hidden/Flat/Gouraud/fLat+edges/gOuraud+edges] <Flat+Edges>: 3d

Command: cone
Specify center point for base of cone or [Elliptical] <0,0,0>: pick any
point Specify radius for base of cone or [Diameter]: 50
Specify height of cone or [Apex]: 70 ENTER



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RESULT:

Thus the 3D view of cone drawing was successfully created using Auto CAD Software.

EX.NO: 18 **3D VIEW OF HEXAGONAL PRISM**
DATE:

AIM: To draw the 3d view of hexagonal prism using Auto CAD Software

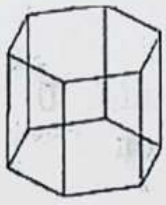
Command: vpoint
Specify a view point or [Rotate] <display compass and tripod>: 1,-
1,1 Regenerating model.

Command: zoom
[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>:
all Regenerating model

Command: shademode
Enter option [2D wireframe/3D
wireframe/Hidden/Flat/Gouraud/fLat+edges/gOuraud+edges] <Flat+Edges>: 3d

Command: polygon
Enter number of sides <4>: 6
Specify center of polygon or [Edge]: pick any point
Enter an option: I
Specify radius of the circle: 35

Command: ext
EXTRUDE
Select objects of extrude: select the object
Specify height of extrusion or [Path]: 70
Specify angle of taper for extrusion <0>: 0 ENTER



RESULT:

Thus the 3D view of hexagonal prism drawing was successfully created using Auto CAD Software.

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EX.NO: 19 **CREATION OF 3-D MODEL OF THE BASE OF MIXER**
DATE :

AIM: To draw the 3d view of base of mixer using Auto CAD Software

Command: vpoint
Specify a view point or [Rotate] <display compass and tripod>: 1,-1,1
Regenerating model.

Command: shademode
Enter option [2D wireframe/3D wireframe/Hidden/Flat/Gouraud/fLat+edges/gOuraud+edges] <Flat+Edges>: 3d

Command: line
Specify first point :pick any point
Specify next point or [Undo]: @0,95,0
Specify next point or [Undo]: @0,-10,65
Specify next point or [Close/Undo]: @0,-30,0
Specify next point or [Close/Undo]: @0,-15,-45
Specify next point or [Close/Undo]: @0,-30,0
Specify next point or [Close/Undo]: c

Command: region
Select objects: Specify opposite corner: 6
found Select objects: all objects
1 loop extracted.
1 Region created

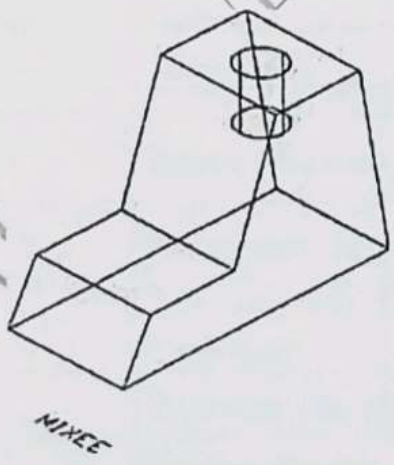
Command: extEXTRUDE
Select objects: select object
Specify height of extrusion or [Path]: 40
Specify angle of taper for extrusion <0>: 0 ENTER

Command: Circle
Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:pick any point
Specify radius of circle or [Diameter] : D
Specify diameter of circle <70.0000>: 15 ENTER

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Command: ext
EXTRUDE
Select objects: select object
Specify height of extrusion or [Path]: 20
Specify angle of taper for extrusion <0>: 0 ENTER

Command: Subtract
Select objects: select the object
Select objects: pick the circle
Select objects: select the object



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RESULT:

Thus the 3D view of base of mixee drawing was successfully created using Auto CAD Software.

ANNEXURE I – FUNCTION KEYS

Function keys	Function defined
F1	Display Help
F2	Command Window on/ off
F3	Object snap on/off
F4	3D osnap on/off
F5	Isoplane top/right/left
F6	Co-ordinate display on/off
F7	Grid on/off
F8	Ortho on/off
F9	Snap on/off
F10	Polar Tracking on/off
F11	Object snap tracking on/off
F12	Toggle dynamic input mode

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ANNEXURE II – CONTROL KEYS

SHORTCUT	COMMAND	FUNCTION
CTRL + 0	Cleanscreen	Turns user interface elements on/off
CTRL + 1	Properties	Turns properties on/off
CTRL + 2	ADCenter	Turns design center on/ off
CTRL + 3	ToolPalettes	Turns tool palettes window on/off
CTRL + 8	Quick Calc	Launches calculator window
CTRL + A		Select All
CTRL + C	Copyclip	Copies objects to clipboard
CTRL + H		Turns a group on or off
CTRL + J		Repeats last command
CTRL + N	New	Opens create new drawing dialogue box
CTRL + O	Open	Opens the select file dialogue box
CTRL + P	Plot	Opens the plot dialogue box
CTRL + R	CVPort	Switches between viewports
CTRL + S	QSAVE	Opens the save drawing dialogue box
CTRL + SHIFT + S		Save as
CTRL + V	Pasteclip	Pastes data from clipboard to drawing
CTRL + X	Cutclip	Removes select object from drawing to clipboard
CTRL + Y	Redo	Performs the operation cancelled by UNDO
CTRL + Z	Undo	Undoes the last operation
CTRL + TAB		Switches between open drawings
CTRL + PAGE UP		Switch up between layout tabs
CTRL + PAGE DOWN		Switch down between layout tabs
ARROW UP/ DOWN		Recall last command

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ANNEXURE III VIVA VOCE QUESTIONS

1. What is meant by AutoCAD?

AutoCAD is one of the cad system software introduced in dec1982 by Autodesk ltd in Sausallia, California of USA. It has hit the technical world like shock wave revolutionizing the manner of drawing preparation.

2. What is the diff between CAD & CADD?

CAD-computer aided design CADD-computer aided design and drafting

3. What are the applications of CAD?

Design of machine elements, CNC machine tools, robotics etc, Panel design and circuit layout Mapping ,building plans, contour plotting and structural drawing Interior design and modeling

4. Define absolute co-ordinates?

Values locating a point in space that describe its displacement from origin (0,0,0)point of the drawing.

5. Define polar co ordinates.

Values are locating a point in space that describes its location relative to the last point picked as defined by an angle and s distance.

6. Define angular dimension?

A dimension that measures the angle between two lines or the angle inscribed by an arc segment

7. Define aligned dimension?

A linear dimension measuring the distance between two points. The dimension line for an aligned dimension is parallel to a line between points.

8. What is Bylayer?

A value that can be assigned to colours and line types so that objects receive their colour and line type properties according to the layer they are on.

9. What is command line?

A text window at the bottom of the screen that's displays command: prompt. This is where you see what you are entering through the keyboard. Also called the

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command window.

10. What is command?

A prompt at the command line, when no commands are currently running.

11. What is cross hair cursor?

A form of the cursor that consists of a horizontal line and a vertical line intersecting at their midpoints, resembling the crosshair in a sighting device.

12. Define dimension variables?

A group of settings and values that control the appreance of dimensions.

13. Define current UCS?

The user co-ordinate system that is active in a drawing. It determines the positive X,Y and Z direction.

14. What is.dwg?

The file extension and format for the standard AUTOcad drawing.

15. Define graphics window?

The appearance of your screen when AUTOcad is running. It consists of the drawing area and surrounding toolbars, menu bars the command window and the status bar.

16. Define line type?

The style of appearance of a line.AUTOcad styles include continues, dashed, dash dot. And so on.

17. What is a hatch patterns?

A pattern of lines dots and other shapes that fills in a closed area.

18. Define MIRROR?

A command that makes a copy of selected objects and flips the copy around a specified line to produce a reciprocal image of those objects.

19. What is an ortho mode?

An on/off setting that, when on forces lines to be drawn and objects to be moved in a horizontal or vertical direction only.

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20. What are the advantages of CAD?

Greater productivity of the designer, improvement of design quality Easier design, calculation and analysis, quicker rate producing drawings, more accuracy of drawings, colour graphics is possible

21. What is an object snap mode?

Any of a set of tools precisely picking strategic points on an object, including end point, midpoint, center, and so on. It is casually called osnap.

22. Define block?

The description of an association of AutoCAD objects that is stored with the drawing file and include a name, an insertion point, and a listening of objects in the association.

23. Define relative co-ordinates?

Values are locating a point in space that describes its displacement from last point picked in the drawing rather than from the origin.

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